CHAPTER 4

Additional Vocations

0916CH04

Mushroom Cultivation

Mushrooms grow from spores, not seeds and develop through a network called mycelium (which acts as 'roots' to absorb nutrients). The essential conditions for mushroom cultivation include proper temperature, humidity, ventilation and hygiene; these conditions can be recreated in moist, dark and cool environments.

Keywords					
	Sterilisation: cleaning and removing germs or other organisms from the growth medium (substrate)	**Mycelium:** root-like network that spreads in the substrate (similar to the mould that develops on the bread)	**Spawn:** a substrate that has growing mycelium in it (acts like a growth initiator for mushroom growth)	**Inoculation:** adding spawn into a larger sterilised substrate	**Incubation:** providing conditions where the inoculated substrate grows (mycelium growing to the entire substrate)

Item/Material name	Description/Use
Spawn	Seed material for mushroom growth
Substrate (Straw/Sawdust/Compost)	Base medium for mushroom growth
Gypsum/Bran/Nutrient Additives	To enrich the substrate
Poly Bags/Trays	For holding the growth medium
Pressure Cooker/Drum	For sterilising substrate
Sprayer/Mist Bottle	To maintain moisture
Thermometer/Hygrometer	For monitoring temperature and humidity
Gloves and Masks	For hygiene during inoculation
Knife/Scissors	For cutting straw and harvesting

Quality parameters

Substrate: Clean, pasteurised straw/sawdust (no odour or mould)

Spawn: White, fresh, uncontaminated mycelium

Environment: Tempeture should be 25–30°C

Humidity: 80–90 per cent; good air flow

Moisture: Substrate damp, not soggy

Harvest: Firm white caps (not flattened or brown)

Storage: Cool at 4–8°C; use ventilated packs

Safety parameters

Hygiene: Clean area, pest-free, no stagnant water

Personal: Gloves, mask, apron; wash hands before handling

Tools: Keep electrical tools dry; handle heaters safely

Chemicals: Use mild disinfectants only.

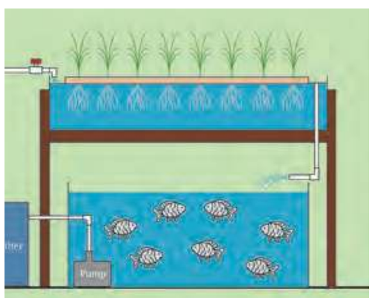
Waste: Compost spent substrate; clean trays regularly

Edibility: Grow only known edible varieties

Key steps in the process

1. **Selection and site design:** Selection of mushrooms as per agro-climatic conditions; design layout for growing area (For example, polybags/trays in shaded area with ventilation and humidity control).
2. **Growth medium preparation:** Testing of growth medium (pH, soil solution, organic carbon, squeeze and mycelium growth) and amendment; sterilisation.
3. **Initiating growth and management:** Preparation and inoculation of spawn; incubation; arrangement in containers.
4. **Management:** Determining irrigation requirements (squeeze test); setting up methods (drip, humidity container); troubleshooting (gaps in sterilisation, maintaining conditions), pest identification and using organic pesticides.
5. **Monitoring growth (physical observation):** Tracking conditions (humidity, temperature, sunlight), observing growth of mushroom (height, colour, change in smell, structure, etc.).
6. **Harvesting, storage and packing:** Identifying when mushrooms are ready for harvesting (caps fully grown, gills visible, firm, etc.); harvesting between flush intervals; different storage methods (paper bags/refrigeration/avoiding plastic bag).

Aquaponics



Aquaponics is a sustainable system that naturally integrates fish (aquaculture) and plant cultivation (hydroponics), making it an efficient method for growing vegetables and raising fish at the same time. In an aquaponics system, plants grow without soil, supported in media beds or floating rafts and thrive in conditions with proper water quality, oxygen, temperature and light.

Keywords			
Keywords	**Nitrogen cycle:** The process where bacteria convert fish waste into nutrients for plants	**Biofilter:** Consists of beneficial bacteria, which purify water	**Flow rate:** Rate at which water moves through the system

Item/Material name	Description/Use
Fish tank	Holds fish and nutrient water
Grow beds	Area where plants grow
Tubing	Moves water through the system
Grow media (gravel/clay pellets)	Supports roots and filters water
Pump	Circulates water to grow beds
Aerator	Adds oxygen for fish
Timer switch	Automates pump cycles
Seeds/Seedlings	Plants for cultivation
Fish feed	Nourishes fish to create nutrients
Water testing kit	Checks pH, ammonia, oxygen and temperature

Safety parameters

Handle with care: Keep electric parts dry
Hygiene: Wash hands before and after work
Protection: Use gloves during cleaning
Dry workspace: Avoid slips and falls
Chemical safety: Be cautious while testing water
Heavy items: Lift tanks or beds safely
Clear labeling: Mark feed and test kits properly
Fish ethics: Avoid overcrowding, handle gently

Quality parameters

Water clarity: Clean and odour-free

pH balance: Slightly acidic to neutral

Plant health: New leaf growth

Fish activity: Active movement, feeding regularly

System flow: Smooth circulation, no blockages

Sustainability check: Minimal water use, no chemicals

Key steps in the process

1. **Designing aquaponics system:** Selection of space; measurement, preparation and assembling materials as per specifications; fabrication of systems (fish tanks, hydroponics system, water pump, timer switch installation and selection of compatible fish–plant pair).
2. **Preparation of grow beds:** Filling with media, that is, gravel or clay pellet, creating bell siphon for filling and draining water to fish tank, testing water quality (pH, nitrate, nitrite, dissolved oxygen)—making amendments.

3. **Initiate growth:** Initiate nitrogen cycle before full fish load (add few fish or ammonia additive), introduce beneficial bacteria (natural colonisation or biofilter media, that is, the gravel or clay pellets); transplant seedlings into grow beds with correct spacing.
4. **Managing the system:** Track water levels and flow rate, recirculation of water, fish feeding based on weight, ensuring proper drainage and overflow control.
5. **Monitoring growth:** Observing and recording plant growth, observing fish behaviour for any disease or stress.
6. **Harvesting, storage and packing:** Harvesting plants as per maturity parameters (change in colour/shape/maturity of leafy vegetables, fruit vegetables and recording yield).

Pisciculture

Fish grow in water systems and their health depends on maintaining the right environmental conditions, such as clean water, appropriate temperature, dissolved oxygen and regular feeding. While fish are naturally farmed in rivers and ponds across many states, suitable growing conditions can also be created in controlled environments.

Key words

Aeration: Adding air or oxygen to the water to help fish stay healthy

Plankton: Tiny plants and animals floating in water that serve as natural food for fish

Spawn: Newly hatched baby fish

Juveniles: Young fish that are older and stronger than spawn

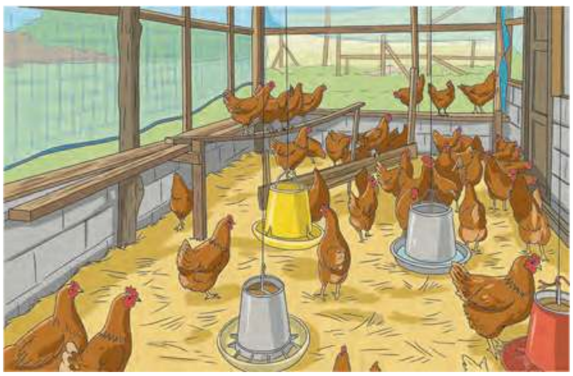
Item/Material name	Description/Use
Pond/tank	Holds fish and nutrient water
Nets	Sampling, catching, and harvesting fish
Feeding trays	Helps monitor feeding and reduces wastage
Buckets	For transporting seed, feed, or harvested fish
Lime and organic manure	Improve pond's soil and water quality
Aerator	Adds oxygen for fish.
Pipes	Manage clean water entry and drainage
Water testing kit	Checks pH, ammonia, oxygen and temperature
Fish feed	Nourishes fish

Quality parameters	Safety parameters
Site prep: Cleaned pond base, apply lime/ manure correctly **Pond:** Clear, no foul odour **Ethical practices:** No injuries to fingerlings. **Water clarity:** Clean and odour-free **Fish activity:** Active movement, feeding regularly **Harvest:** No deformity or signs of disease **Storage:** Kept cool, handles hygienically	**Protection:** Use gloves, non-slippery footwear **Chemical safety:** Be cautious while testing water, apply lime/manure in correct dose **Fish ethics:** Avoid overcrowding, handle gently **Equipment safety:** Maintain aerators and electrical lines to prevent shocks or accidents

Key steps in the process

1. **Layout:** Selection of site, measuring and assembling materials, removal of weeds, ensuring proper inlet/outlet.
2. **Preparation of fish pond:** Filling clean water, testing water quality (pH, dissolved oxygen, temperature, ammonia, nitrite), making amendments.
3. **Initiate growth:** Add cow dung slurry/manure before introducing fish to stimulate natural plankton (starter food), introducing spawn or older juveniles into the pond based on size.
4. **Managing the system:** Feed rice bran, vegetables like kale, spinach, feed twice daily, install feeding trays for observation, avoid overfeeding (leads to ammonia), remove debris/fallen litter.
5. **Monitoring growth:** Take samples using nets every 2 weeks for observation, measure average weight, adjust feed accordingly; observe fish behaviour, feeding response, and signs of stress or disease.
6. **Harvesting, storage and packing:** Observing matured fish (size of fish as per marketable size), harvesting using net or draining of water, storage in refrigerator, sorting as per size and quality.

Backyard Poultry



Poultry farming does not necessarily need large spaces, instead it can be managed in small spaces like home courtyards, verandas or simple sheds made from locally-available materials. Well-being of poultry depends on suitable housing, balanced feed, clean drinking water and protection from predators. The essential conditions for successful poultry rearing include proper ventilation, warmth for young chicks, hygiene of the coop and regular health checks.

Keywords	Coop	Brooder	Feeding response	E. coli	Wet bedding
	A small shelter or house where chickens are kept safe and protected	Special warm house for very young chicks	How actively and eagerly chicks eat their food, showing their health	Bacteria found in contaminated water and causes sickness in humans and chickens	Damp litter on the floor that must be removed to prevent disease

Item/Material name	Description/Use
Nesting boxes	Space for hens to lay clean, protected eggs
Bedding material	Keeps the coop dry and comfortable
Fencing/Mesh	Prevents predators and keeps birds contained
Feed (grains, mash, kitchen greens)	Nutrition for growth and egg production
Cleaning tools	Maintain hygiene inside the coop
Health kit	For cleaning, parasite control and minor illnesses
Lighting bulb/Solar lamp	Provides light for safety and improves laying of eggs in winter

Quality parameters
Bird health: Active behaviour, clean eyes, good appetite, normal droppings
Egg quality: Clean shells, good shape, consistent size
Feed quality: Fresh, dry, balanced feed; no mould
Water quality: Clean, changed daily
Housing quality: Dry, ventilated, predator-proof coop
Hygiene: Regular cleaning, dry bedding, low ammonia smell
Security: Limited outsider entry, separate sick birds

Safety parameters
Coop safety: Strong fencing, secure doors, raised floor
Bird safety: Gentle handling, separation of aggressive or sick birds
Feed safety: Store feed in closed containers; avoid spoiled or contaminated feed
Cleanliness: Regular coop cleaning to prevent parasites and infections
Water safety: Provide clean water; prevent contamination from droppings
Environmental safety: Safe disposal of waste and dead birds; maintain good ventilation

Key steps in the process

1. **Layout:** Selection of site (clean, dry, removal of debris), for coop-elevation (if possible) to avoid water logging, using jute sac to build walls, litter layer with rice husk/sawdust on the floor as bedding, for brooder-cardboard or wooden box, electric bulb, bowl for water, sawdust, etc.

2. **Preparing the coop:** Access to fresh water and feed, set up of poultry shed (proper ventilation, sunlight, barriers from predators), water testing (for E.coli infection).
3. **Initiate growth:** Introducing clean and disinfected chicks in the coop, maintain temperature conditions.
4. **Managing the coup:** Regular cleaning, replacing drinking water, remove wet bedding, feeding the chicks as per requirement.
5. **Monitoring growth:** Observing chick activity, feeding response, signs of illness, egg laying per bird, etc.
6. **Harvesting, storage and packing:** Collecting eggs gently and storing in cool place, handling mature chickens safely for transport who have reached the desired height, etc.

Non-Timber Forest Produce

Non-Timber Forest Produce (NTFP) includes a wide variety of natural products, such as honey, medicinal plants, resins, gums, fruits, seeds, leaves and bamboo–items that support nutrition, health and livelihoods without cutting down trees. While most NTFP collection happens in natural forest conditions, processing steps, such as drying, sorting or extraction can easily be carried out in community spaces.

Keywords

Sustainable: Using natural resources in a careful way so they can continue to grow and be available for future use

Lac: A natural resin produced by a tiny insect that lives on certain trees (ber, kusum). It hardens on the branches and is collected to make products like varnish, toys, dyes and polishes.

Mulching: Covering the soil with materials, like dry leaves, straw, grass or plastic sheets to protect it and help plants grow better.

Item/Material name	Description/Use
Collection basket/Bag	Gather leaves, seeds, fruits, gums without damage
Sickle/Hand pruner	Cut grasses, twigs and mature leaves
Knife/Blade	Trim bark, scrape gums/resins
Gloves	Protect hands during harvesting and handling
Storage jars/Containers	Store dried NTFPs and keep moisture out
Sieves	Remove dirt and impurities from collected material

Quality parameters	Safety parameters
Collection tools: Clean, sharp, safe, non-contaminated **Suitable collection site:** Unpolluted forest area, no chemical exposure **Right maturity stage:** Leaves, seeds, gums, resins, or flowers at proper maturity **Sustainable harvesting:** No uprooting; partial collection; avoid damaging host trees **Clean handling:** Remove debris, avoid contamination, use gloves	**Personal safety:** Use gloves and proper footwear **Tool safety:** Handle sickles/knives carefully, store safely **Environmental safety:** Avoid disturbing wildlife **Fire safety:** Keep drying areas away from open flames and extremely dry plant materials

Key steps in the process

1. **Selection of area:** Identification of NTFP, selection of safe and accessible forest area for collection; with sustainable practices and minimal disturbance.
2. **Maintaining growth medium of NTFP:** Soil testing (texture, moisture), gathering meteorological data (climate), sunlight availability, etc.
3. **Management of the NTFP/Host to NTFP:** Cleaning around the area, removing plastic waste/spoiled produce, protection from human activities, mulching or adding leaf–litter compost.
4. **Monitoring of the NTFP/Host to NTFP:** Observing and tracking health, checking for signs of stress, pests, stage of maturity, etc.
5. **Harvesting, packing and storage:** Safe use of tools to harvest targeted parts of the NTFP (leaves, seeds, twigs for lac, etc.), cleaning of the harvest (sorting, removing dirt, drying), storage (cloth, tie in bundles) and packaging (airtight, plastic or carboard box).